

# Data Science and Machine Learning for Geoscientists

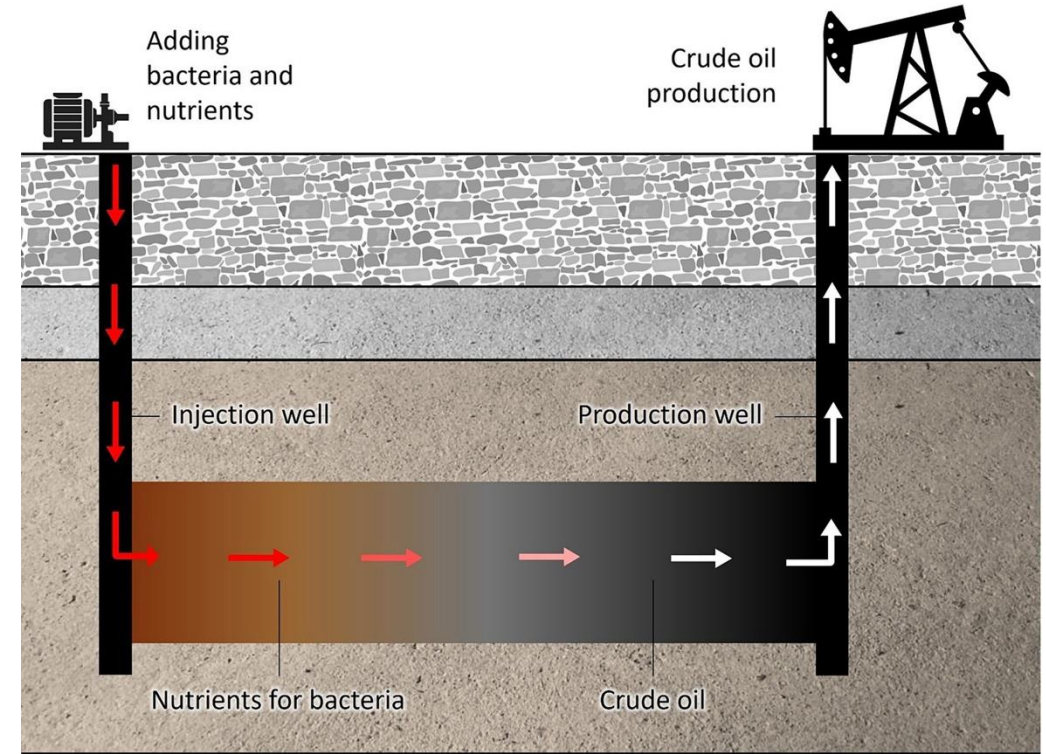
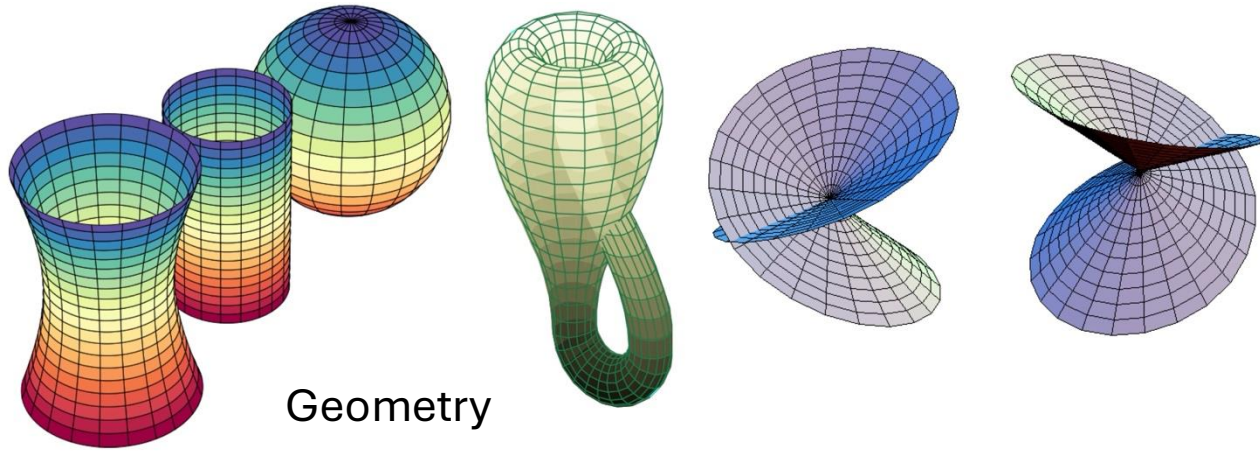
## *Lecture 1: Geoscience Data and Getting Started with Python*

Samaneh Abolpour Mofrad

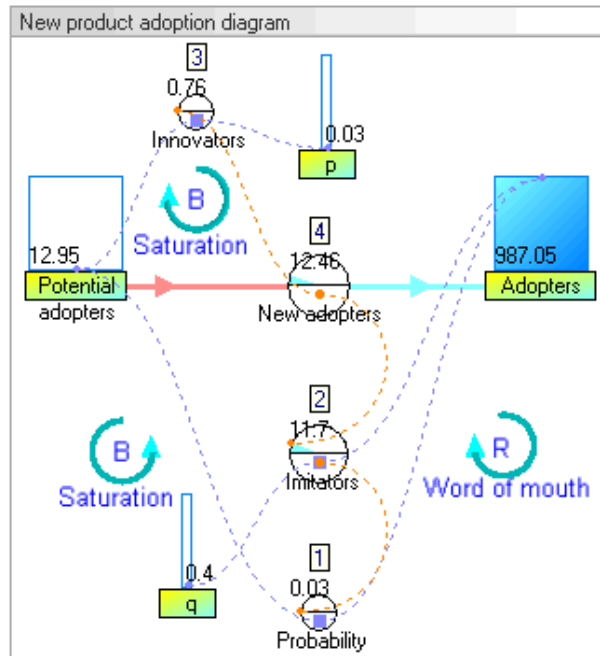
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8 October 2026

# About Me



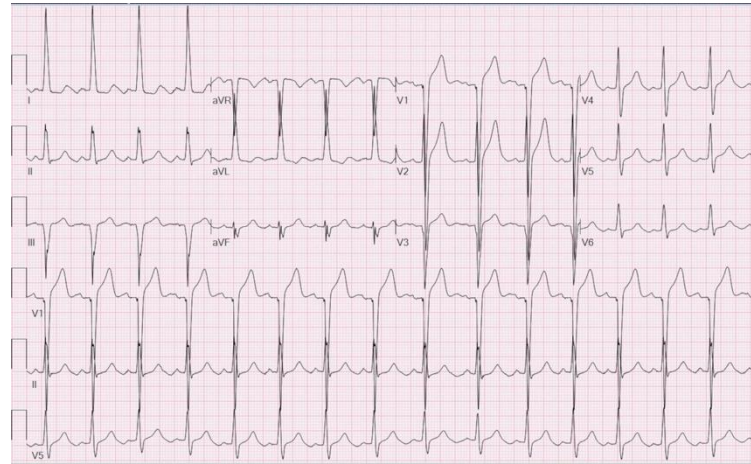
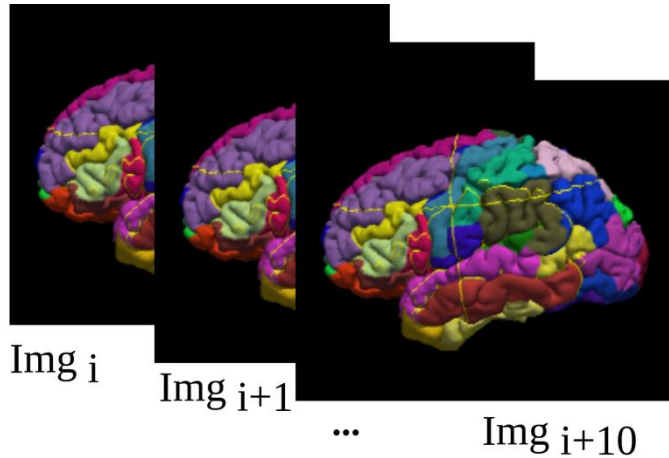
Porous Media



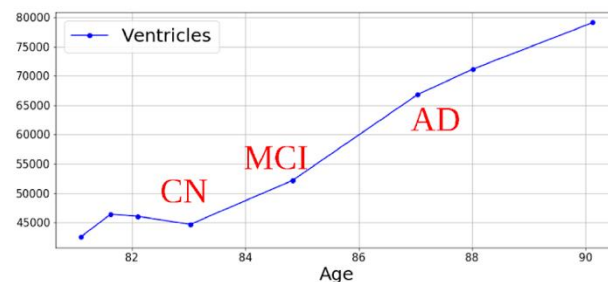
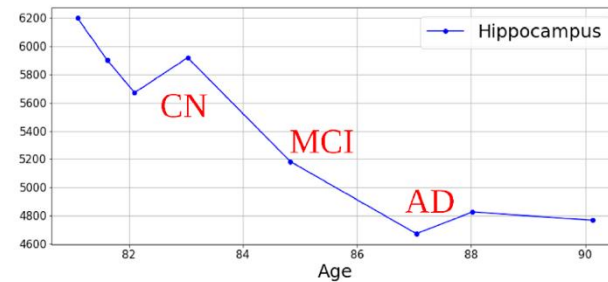
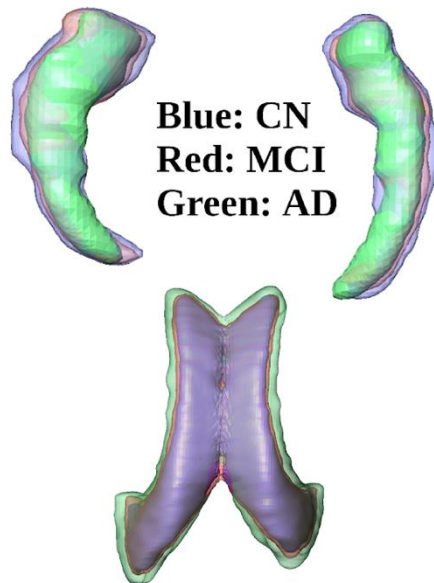
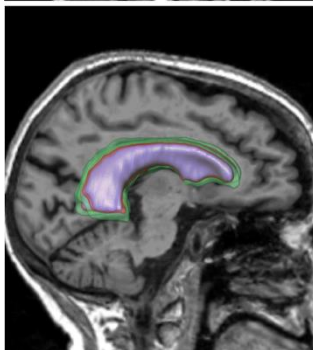
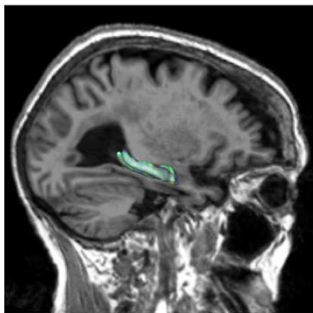
System Dynamics

# Examples from my research

Predicting the risk of dementia  
based on longitudinal data



Texas Children's Hospital: Initiated development of a deep-learning model to classify paediatric ECG signals.

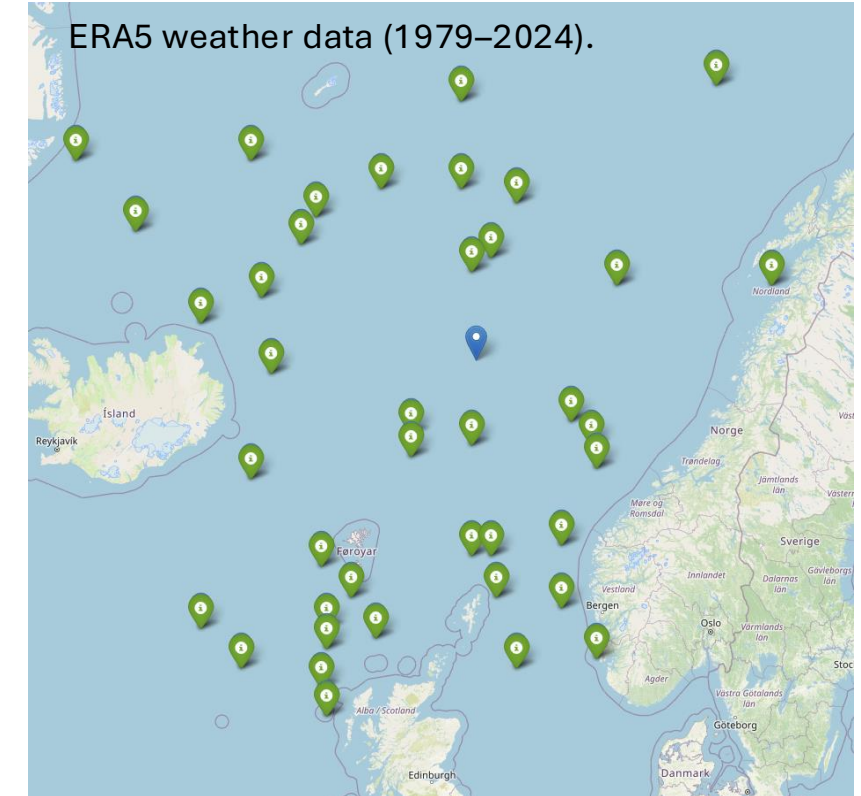
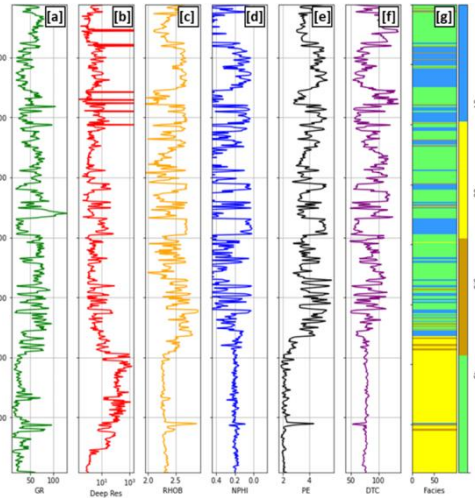


Drug repurposing for Parkinson's disease



# Examples from my research

Aker BP (oil & gas): anomaly detection/classification to detect faulty drilling logs



Time-series forecasting: weather, wind power, price

Image analysis: object detection, classification, location estimation and similarity search.



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# Course outcomes

## What you will be able to do by the end of the course?

- Turn a geoscience question into a data-analysis task.
- Explore and prepare geoscience data using Python.
- Build and tune machine-learning models.
- Evaluate models and interpret their results.
- Document an analysis that someone else can follow.

A typical output: a notebook with a question, data checks, plots, a model evaluation and a scientific conclusion.

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# Course roadmap

| Foundations                  | Modelling and interpretation                |
|------------------------------|---------------------------------------------|
| Python and Jupyter notebooks | Linear models                               |
| NumPy and Pandas             | Model selection and tuning                  |
| Exploratory data analysis    | Regularisation and dimensionality reduction |
| Data preparation             | Introduction to deep learning               |

# Learning and assessment

## Classes:

- Short explanations, self-paced notebooks and videos.
- Course materials and videos shared through GitHub.
- Try problems, compare approaches and discuss solutions.
- Support from me and the GTAs while you work.

*Quick check: Python is new to me / I have used it a little / I use it regularly.*

## Assessment:

Not yet confirm but two options are considered:

1. Only a final exam, like last year an offline exam.
2. Assignment + final exam: one third or half for the assignment, the remainder for the final exam.
3. None of the coding that you do in the lectures is assessed but do your best to learn from the exercises.

Do you have any preference?

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# Today's Session

| Time        | Focus                                           |
|-------------|-------------------------------------------------|
| 09:00–09:20 | Course aims, learning and assessment discussion |
| 09:20–09:30 | Geoscience data and questions                   |
| 09:30–10:15 | Python tools, and notebook access               |
| 10:15–10:30 | Break                                           |
| 10:30–11:50 | Supported Python practice                       |
| 11:50–12:00 | Review and next steps                           |

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# Geoscience Data

Geoscience data refers to any quantitative, qualitative, or spatial information collected, measured, or modeled about the Earth's physical structure, composition, history, and natural processes ([source](#)).

- Geoscience data can come from a rock sample, a sensor, a field survey or an observation made from space. We may also work with outputs from models.
- Question: what does this value describe, and where and when does it apply?

# Geoscience Data

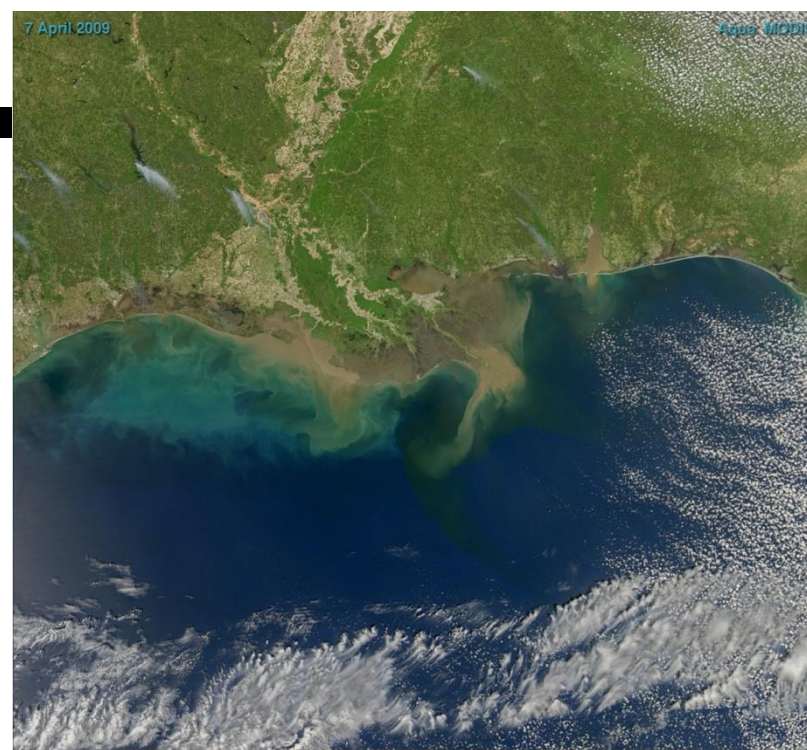
| Data in the course | Example information                          |
|--------------------|----------------------------------------------|
| Mineral properties | Name, density, chemical family               |
| Sediment age-depth | Depth and age markers                        |
| Core measurements  | Porosity, gamma radiation, colour            |
| Geochemistry       | Calcium carbonate content                    |
| Borehole logs      | Gamma radiation, resistivity, lithology      |
| Earthquakes        | Event locations; roadside habitat attributes |

What (other) types of data have you worked with that are missing from this list?

Sources: BGS data collections and course notebooks.

# Sediments and boreholes

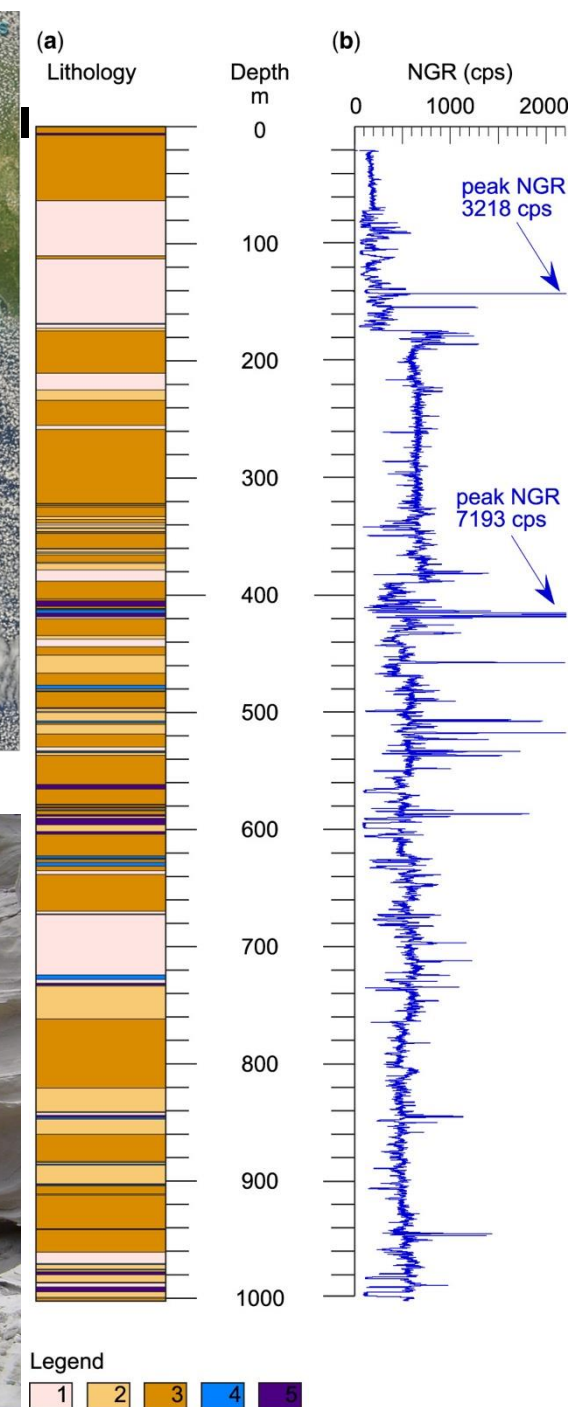
- In earth and planetary sciences, [sediment](#) is a solid material made of loose particles that is transported to a new location most often with water, wind, and glaciers.
- Age-depth markers: how does sediment age change with depth?
- Borehole logs record measurements or observations against depth, helping us describe subsurface materials.



Sediment in the [Gulf of Mexico](#)



Glacial sediments from Montana



# Mineral data

- Mineral data can describe a mineral's name, density and chemical group.
- Numerical data: density, measured in  $\text{g/cm}^3$ .  
Categorical data: mineral name and group, such as carbonate or silicate.



[Source: Data-driven discovery reveals Earth's missing minerals](#)



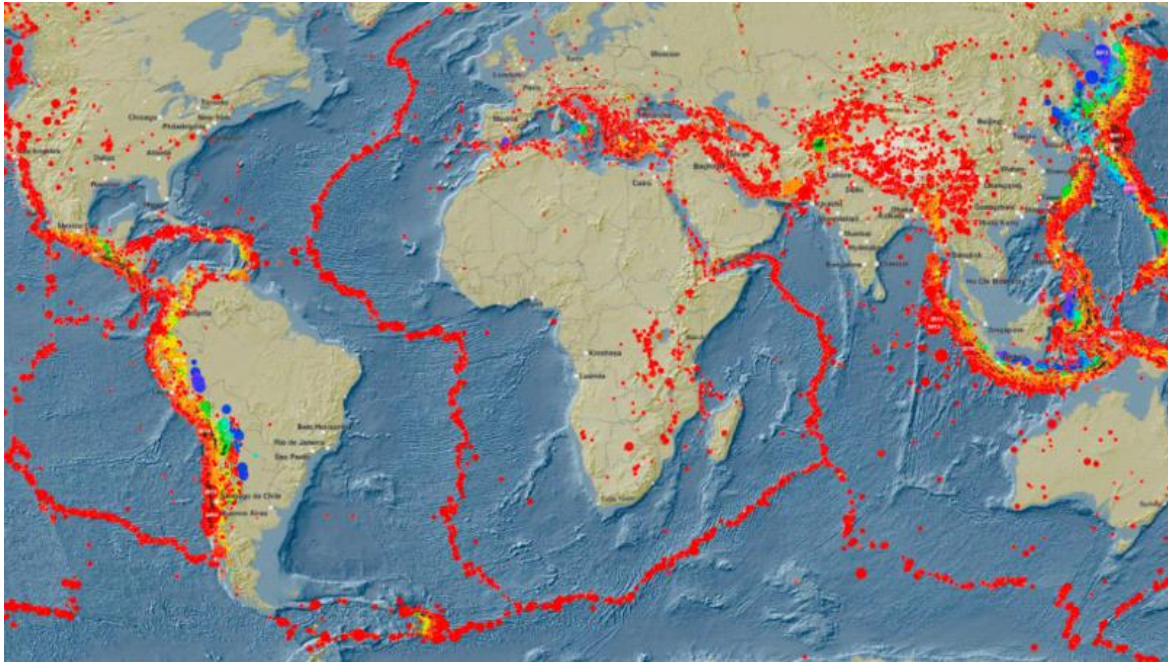
# Satellite observations



- Lake Mead: Landsat observations, 1985–2025
- A sequence of images can show change through time.
- Satellite observations help us monitor changes in weather, oceans and land surfaces.

Lake Mead has the largest storage capacity of any reservoir in the U.S. Decades of increasing water demand and drought in the American West have caused a continual decline in water levels ([source](#)).

# Earthquake data



[Source](#)

World map of earthquakes – showing all earthquakes recorded by the GEOFON network between 2008 and 2023. Shallow earthquakes (up to 50 km deep) are marked in red. All other colour codes indicate earthquakes at greater depths.

- Analyse patterns in location and magnitude
- An earthquake catalogue records event locations, depths and magnitudes.
- Maps and plots help us explore where recorded earthquakes occur and how their properties vary.
- In this course, we use California earthquake data for exploratory analysis.

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# Exploring data before modelling

- Exploratory data analysis (EDA) helps us understand what our data contain.
- What does each row or measurement represent?
- Are units, missing values and unusual observations understood?
- What do plots show about variation and relationships?
- Do we have the data needed for our question?

# Python tools

| Tool                 | What we use it for                     |
|----------------------|----------------------------------------|
| NumPy                | Calculations with numerical arrays     |
| pandas               | Tables, filtering and data preparation |
| Matplotlib / seaborn | Plots and exploratory visualisation    |
| scikit-learn         | Machine-learning models and evaluation |

- JupyterLab / Jupyter Notebook: where we combine code, results and explanations.

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# Use AI to Support Your Thinking

- Try first: attempt the problem and identify where you are stuck.
- Ask for support: request a hint, an explanation or feedback on your reasoning. “Here is my attempt and the error message. Give me one hint at a time so I can find the problem myself.”
- Be able to explain the code you use and interpret its results.
- **Be curious. Ask questions. Make mistakes. Give yourself the chance to learn.**
- Use of AI will **NOT** be allowed in your assessed coursework